

## EDWARDS COUNTY AP, TX, USA

WMO: 722363

Lat: 29.947N

Lon: 100.173W

Elev: 2372

StdP: 13.48

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 23098

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 27.8       | 31.6 | 9.0                           | 9.5 | 43.7 | 11.8 | 10.9 | 45.9 | 23.8                  | 51.0 | 20.5 | 53.2 | 4.7                   | 30   | 0.458 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 21.1                   | 97.3            | 70.1 | 95.2 | 69.9 | 93.0 | 70.2 | 75.4                | 84.9 | 74.7 | 84.0 | 73.9 | 83.0 | 9.9                  | 150  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 73.0                            | 133.9 | 77.8 | 72.4 | 131.2 | 77.4 | 72.1 | 129.6 | 77.2 | 40.6          | 86.1 | 39.9 | 83.0 | 39.2 | 82.8 | 86.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| 20.8              | 18.8 | 17.4 | DB                         | 23.3 | 100.2              | 2.9 | 2.7                                                | 21.3 | 102.2      | 19.6 | 103.8      | 17.9 | 105.3      | 15.9 | 107.3 | (4) |
|                   |      |      | WB                         | 20.0 | 77.6               | 3.0 | 2.8                                                | 17.8 | 79.6       | 16.1 | 81.2       | 14.4 | 82.8       | 12.2 | 84.8  | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 67.1   | 49.8  | 54.9  | 60.6  | 67.0  | 73.4  | 80.2  | 81.4  | 82.1  | 77.1  | 68.5  | 58.3  | 51.3 |
|                                                                                  | DBStd       | 13.36  | 9.00  | 9.43  | 8.41  | 6.40  | 5.77  | 3.52  | 3.24  | 3.09  | 4.55  | 7.54  | 8.75  | 8.70 |
|                                                                                  | HDD50       | 319    | 115   | 54    | 20    | 2     | 0     | 0     | 0     | 0     | 0     | 4     | 31    | 93   |
|                                                                                  | HDD65       | 1717   | 473   | 295   | 180   | 50    | 8     | 0     | 0     | 0     | 1     | 54    | 227   | 429  |
|                                                                                  | CDD50       | 6561   | 109   | 191   | 347   | 513   | 724   | 905   | 974   | 996   | 814   | 577   | 279   | 132  |
|                                                                                  | CDD65       | 2488   | 3     | 12    | 43    | 112   | 268   | 455   | 509   | 531   | 364   | 162   | 25    | 4    |
|                                                                                  | CDH74       | 24040  | 30    | 127   | 315   | 849   | 2255  | 4727  | 5459  | 5942  | 3117  | 1065  | 122   | 32   |
|                                                                                  | CDH80       | 10944  | 2     | 24    | 62    | 241   | 872   | 2293  | 2761  | 3091  | 1306  | 276   | 13    | 3    |
| Wind                                                                             | WSAvg       | 8.7    | 8.0   | 8.4   | 9.8   | 10.3  | 9.9   | 10.2  | 8.9   | 8.1   | 7.1   | 8.2   | 7.9   | 8.1  |
| Precipitation                                                                    | PrecAvg     | 25.9   | 1.1   | 1.1   | 1.9   | 1.9   | 3.0   | 2.8   | 2.6   | 2.6   | 3.3   | 2.7   | 1.8   | 1.0  |
|                                                                                  | PrecMax     | 43.5   | 2.7   | 3.3   | 6.4   | 5.9   | 6.7   | 8.1   | 9.3   | 15.4  | 8.9   | 9.2   | 7.8   | 5.6  |
|                                                                                  | PrecMin     | 11.7   | 0.0   | 0.0   | 0.0   | 0.0   | 0.8   | 0.0   | 0.2   | 0.3   | 0.4   | 0.0   | 0.0   | 0.1  |
|                                                                                  | PrecStd     | 8.3    | 0.9   | 1.1   | 1.5   | 1.5   | 1.6   | 2.4   | 2.6   | 3.1   | 2.0   | 2.5   | 2.2   | 1.2  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 78.7  | 83.6  | 84.4  | 91.1  | 95.4  | 99.4  | 99.1  | 99.4  | 97.3  | 90.0  | 81.6  | 78.8 |
|                                                                                  |             | MCWB   | 55.3  | 58.7  | 58.7  | 60.4  | 67.1  | 69.1  | 70.2  | 70.0  | 68.2  | 67.7  | 63.0  | 55.2 |
|                                                                                  | 2%          | DB     | 72.6  | 78.7  | 81.5  | 86.0  | 90.9  | 96.5  | 97.0  | 97.2  | 92.6  | 85.9  | 77.5  | 72.7 |
|                                                                                  |             | MCWB   | 54.8  | 56.2  | 59.8  | 61.5  | 68.5  | 70.0  | 70.0  | 70.7  | 69.2  | 68.4  | 61.9  | 57.0 |
|                                                                                  | 5%          | DB     | 67.8  | 73.0  | 78.5  | 81.9  | 88.4  | 92.9  | 93.5  | 95.1  | 90.2  | 82.4  | 73.3  | 68.3 |
|                                                                                  |             | MCWB   | 54.4  | 56.4  | 60.0  | 62.6  | 68.4  | 70.2  | 70.6  | 70.4  | 69.3  | 67.1  | 60.9  | 55.9 |
|                                                                                  | 10%         | DB     | 63.6  | 69.6  | 73.0  | 79.2  | 84.3  | 90.7  | 91.4  | 92.8  | 87.9  | 80.9  | 71.6  | 64.0 |
|                                                                                  |             | MCWB   | 53.0  | 56.5  | 59.2  | 62.5  | 68.1  | 70.4  | 70.3  | 70.5  | 69.0  | 66.3  | 60.1  | 55.9 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 63.7  | 65.8  | 67.8  | 71.2  | 75.4  | 77.5  | 76.3  | 75.8  | 75.3  | 75.0  | 70.0  | 65.6 |
|                                                                                  |             | MCDB   | 65.7  | 71.1  | 74.9  | 78.2  | 83.6  | 88.2  | 87.5  | 87.0  | 84.7  | 81.0  | 74.1  | 68.1 |
|                                                                                  | 2%          | WB     | 61.3  | 63.7  | 66.4  | 69.8  | 73.7  | 75.6  | 74.9  | 74.9  | 74.1  | 72.6  | 67.1  | 63.6 |
|                                                                                  |             | MCDB   | 65.7  | 66.4  | 72.3  | 76.1  | 81.8  | 84.6  | 84.7  | 85.4  | 82.5  | 78.7  | 70.9  | 65.3 |
|                                                                                  | 5%          | WB     | 58.5  | 62.2  | 64.8  | 68.3  | 72.5  | 74.5  | 74.0  | 74.1  | 73.2  | 71.0  | 65.3  | 60.9 |
|                                                                                  |             | MCDB   | 64.0  | 66.0  | 69.6  | 74.8  | 80.5  | 83.1  | 83.5  | 84.5  | 81.1  | 77.4  | 69.0  | 64.4 |
|                                                                                  | 10%         | WB     | 55.0  | 59.8  | 63.4  | 67.0  | 71.5  | 73.3  | 73.2  | 73.2  | 72.3  | 69.7  | 63.4  | 57.8 |
|                                                                                  |             | MCDB   | 60.9  | 66.0  | 67.5  | 73.4  | 79.3  | 81.8  | 82.6  | 83.7  | 80.2  | 75.9  | 67.1  | 62.2 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 19.7  | 20.5  | 20.5  | 20.8  | 19.7  | 20.5  | 20.4  | 21.1  | 18.9  | 19.4  | 19.1  | 19.5 |
|                                                                                  |             | MCDBR  | 24.4  | 25.0  | 24.8  | 24.7  | 24.1  | 23.9  | 23.2  | 24.1  | 22.2  | 20.3  | 20.9  | 23.8 |
|                                                                                  |             | MCWBR  | 13.2  | 12.5  | 11.3  | 11.7  | 9.0   | 6.5   | 6.4   | 6.5   | 7.3   | 9.2   | 10.9  | 13.1 |
|                                                                                  | 5% WB       | MCDBR  | 19.6  | 18.1  | 17.6  | 17.5  | 18.5  | 18.9  | 19.5  | 20.3  | 17.1  | 17.0  | 17.2  | 17.6 |
|                                                                                  |             | MCWBR  | 14.4  | 12.8  | 10.0  | 9.7   | 8.4   | 7.7   | 6.0   | 6.2   | 6.2   | 9.1   | 11.9  | 13.7 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.293  | 0.298 | 0.319 | 0.349 | 0.373 | 0.378 | 0.390 | 0.376 | 0.366 | 0.335 | 0.311 | 0.295 |      |
|                                                                                  | taud        | 2.568  | 2.558 | 2.510 | 2.441 | 2.426 | 2.463 | 2.473 | 2.500 | 2.462 | 2.541 | 2.545 | 2.564 |      |
|                                                                                  | Ebn at Noon | 302    | 309   | 307   | 299   | 291   | 287   | 283   | 287   | 288   | 292   | 292   | 296   |      |
|                                                                                  | Edh at Noon | 26     | 29    | 33    | 36    | 37    | 36    | 35    | 34    | 34    | 30    | 27    | 25    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1024   | 1243  | 1556  | 1840  | 1916  | 2172  | 2154  | 2036  | 1669  | 1365  | 1070  | 922   |      |
|                                                                                  | RadStd      | 115    | 180   | 144   | 169   | 204   | 172   | 180   | 113   | 147   | 196   | 140   | 123   |      |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) | Regional (20 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | +209  | +157  |

Nomenclature: See separate page